

RESEARCH INTERESTS

Computational approaches to optimal transport and Wasserstein distributionally robust optimization problems.

EMPLOYMENT

- 2024 - now **Assistant Professor**
College of Engineering & Computer Science, VinUniversity
Vietnam
- 2023 - 2024 **Research Fellow**
Department of Mathematics, National University of Singapore
Singapore
- 2017 - 2018 **Research Assistant**
Institute of Mathematics, Vietnam Academy of Science and Technology
Vietnam

EDUCATION

- 2018 - 2023 **Ph.D**
Department of Mathematics, National University of Singapore
Singapore
(Supervisor: Professor Kim-Chuan Toh)
- 2013 - 2017 **Bachelor of Mathematics**
Faculty of Mathematics & Information Technology, Hanoi National University of Education,
Vietnam

PUBLICATIONS & PREPRINTS

- 2026 **On the Convergence of Wasserstein Gradient Descent for Sampling**
Van Chien Ta, Hong T.M. Chu, Minh-Ngoc Tran
Submitted.
- 2026 **Concave Certificates: Geometric Framework for Distributionally Robust Risk and Complexity Analysis**
Hong T.M. Chu
Submitted.
- 2025 **Time-integrated Optimal Transport: A Robust Minimax Framework**
Thai P. D. Nguyen, Hong T.M. Chu, Kim-Chuan Toh
Submitted.
- 2025 **Tight Robustness Certificates and Wasserstein Distributional Attacks for Deep Neural Networks**
Bach C. Le, Tung V. Dao, Binh T. Nguyen, Hong T.M. Chu
Submitted.
- 2023 **Wasserstein distributionally robust optimization and its tractable regularization formulations**
Hong T.M. Chu, Meixia Lin, Kim-Chuan Toh.
Journal of Optimization Theory and Applications 208, 69 (2026).
- 2023 **A Corrected Inexact Proximal Augmented Lagrangian Method with a Relative Error Criterion for a Class of Group-quadratic Regularized Optimal Transport Problems**
Lei Yang, Ling Liang, Hong T.M. Chu, Kim-Chuan Toh.
Journal of Scientific Computing 99 (3), 79.

2021

On Regularized Square-root Regression Problems: Distributionally Robust Interpretation and Fast Computations
Hong T.M. Chu, Kim-Chuan Toh, Yangjing Zhang.
Journal of Machine Learning Research, 23(308):1–39.

2020

An efficient implementable inexact entropic proximal point algorithm for a class of linear programming problems
Hong T. M. Chu, Ling Liang, Kim-Chuan Toh, Lei Yang.
Computational Optimization and Applications, (2023): 1-40.

HONORS AND AWARDS

2025

Teaching award
Full-time teaching faculty award AY 2024/2025
VinUniversity, Vietnam

2020, 2021

Teaching award
Part-time teaching assistant award AY 2019/2020, 2020/2021
Faculty of Science, National University of Singapore, Singapore

ADMINISTRATIVE EXPERIENCE

July 2026

Organizing Committee
Mini-Course: Variational Analysis and Distributionally Robust Optimization in Learning, VIASM Vietnam

July 2026

Organizing Committee
Learning and Optimization for a Sustainable Future, VinUniversity, Vietnam

2023 - 2028

Ambassador
Appointment as Faculty of Science Postgraduate Alumni Ambassador (FPAA)
National University of Singapore, Singapore

Spring 2024

Preliminary & Final Judge
Singapore Science & Engineering Fair (SSEF)
MoE, A*STAR and SCS, Singapore

Jan 2024

Member of Organizing Committee
The Mathematics of Data, Institute for Mathematical Sciences, Singapore

REVIEWER SERVICE

JOTA, EJOR, ICML 26, ACML 24, AISTATS 25/26, OIC 24/25, SOICT 25

PEDAGOGICAL EXPERIENCE

Spring 2024, Spring 2025	Instructor, DATA1010 Introduction to Data Science (Vin-University) <i>Average student rating: 4.70/5.0</i>	Spring 2020	Teaching assistant, MA3252 Linear and Network Optimisation (National University of Singapore) <i>Average student rating: 4.4/5.0</i>
2024-2025, 2025-2026	Instructor, MATH2010&MATH4010 (Advanced) Probability & Statistics (VinUniversity) <i>Average student rating: 4.50/5.0</i>	Fall 2019	Teaching assistant, MA1102R Calculus (National University of Singapore) <i>Average student rating: 4.4/5.0</i>
Spring 2022, Fall 2021, Fall 2020	Teaching assistant, DSA3102 Essential Data Analytics Tools: Convex Optimisation (National University of Singapore) <i>Average student rating: 4.1/5.0</i>		

PROGRAMMING LANGUAGES

MATLAB, PYTHON